

QRC Usefulness Atlas: Visual Abstract

Outcome composition, family structure, anti-circularity robustness, and quantum-attribution guardrails in one reproducible view.

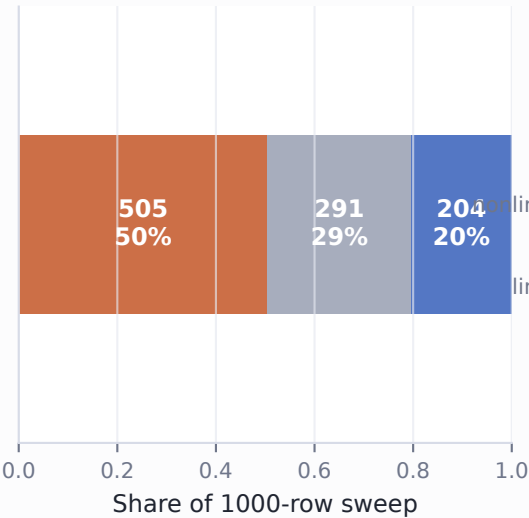
ATLAS ROWS
1,000
50 generators

QRC-USEFUL
204
20.4% of sweep

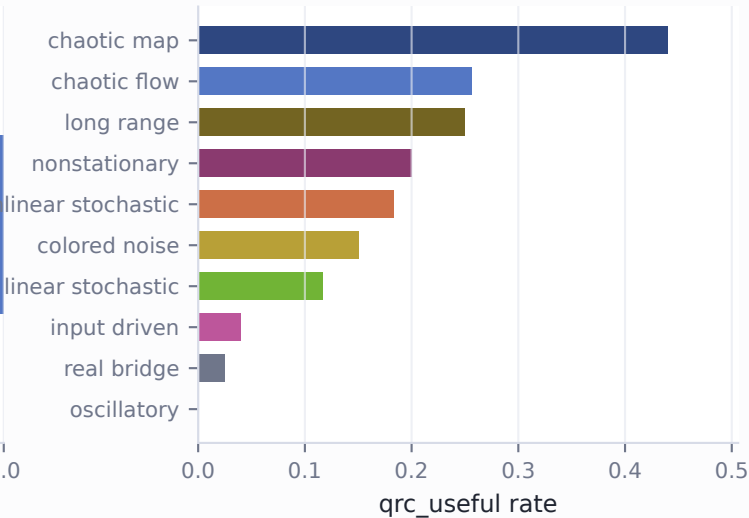
META-MODEL
R2 0.67
AUC 0.86

NO-PROXY CHECK
R2 0.66
AUC 0.86

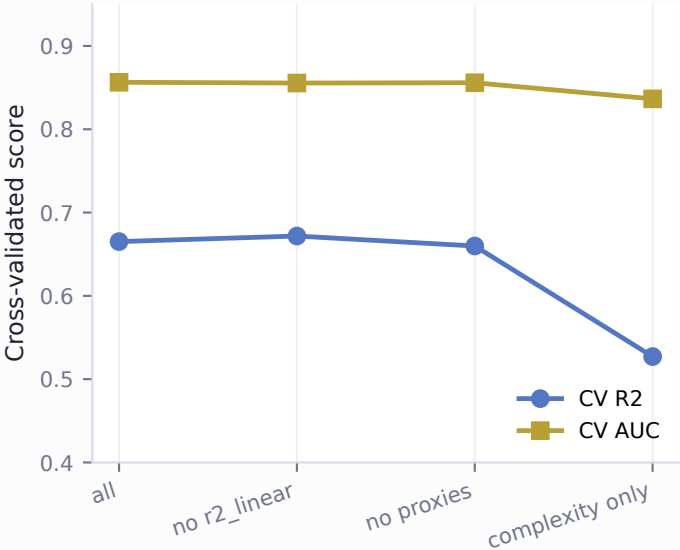
Outcome composition



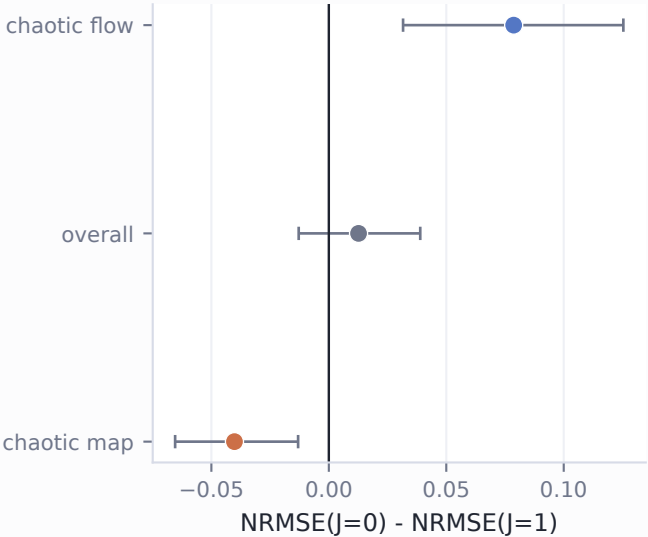
Family useful-rate ranking



Anti-circularity feature-set checks



Paired coupling-control intervals



Main claim: QRC usefulness is dataset-conditional and learnable from measured properties.

Strongest family: chaotic map: useful rate 44.0%.

Guardrail: Overall $J=1$ vs $J=0$ CI [-0.013, 0.039] crosses zero.

Not claimed: No broad average QRC superiority and no coupling/entanglement mechanism claim.